Yongzhuo Chen

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Academic Profile

My research focuses on developing statistical and computational methods to understand complex biological systems through *principled information integration across time, cell types, and heterogeneous datasets.* My methodological research combines *high-dimensional inference*, *adaptive transfer learning*, and *genomic foundation models*, with applications to dynamic gene-expression networks, human brain development, and neurological disease genetics.

Research Interests

Statistical inference; Adaptive transfer and multi-task learning; Statistical genomics; Genomic foundation models.

Education

University of Florida — *Expected May 2027*
Ph.D. in Biostatistics — Gainesville, Florida
Research advisors: Drs. Muxuan Liang and Sai Zhang

Yale University — *2022–2024*
Master of Science in Biostatistics (GPA: 3.75) — New Haven, Connecticut

Huazhong University of Science and Technology — *2018–2022*
Bachelor of Science in Physics (GPA: 3.88) — Wuhan, China

Honors & Awards

2025 Distinction Award for the Written Qualifying Examination — *2026*
Department of Biostatistics, University of Florida; ranked first out of 12 .

Publications & Preprints

[1] Sohom Bhattacharya†, **Yongzhuo Chen***, and Muxuan Liang† (2025). *Late Fusion Multi-task Learning for Semiparametric Inference with Nuisance Parameters*. arXiv:2507.07941. **Accepted for publication** in the *Journal of the American Statistical Association* (Theory and Methods).
** Lead/first author (author list alphabetical); † Co-corresponding authors.*

[2] Bonsall, S., Kazu, R. S., King, M., Leung, D., Mahiddine, F. Y., Highley, J. R., Strange, A., **Chen, Y.**, et al. (2026). *Single-nucleus multiomic atlas of ALS primary motor cortex nominates neuroprotective WDR49-expressing astrocytes*. Research Square preprint. Under revision at *Nature Neuroscience*.

[3] **Yongzhuo Chen**, Yixuan Liang, Yiran Liu, Brian Hobbs, and Michael Kane (2023). *Valuing Post-Revenue Biopharmaceutical Assets with Pfizer's Current Portfolio as a Case Study*. arXiv:2312.02250.

Manuscripts in Preparation

[1] **Yongzhuo Chen**, Sohom Bhattacharya, and Muxuan Liang. *Adaptive Estimation and Inference for Time-Varying Transition and Precision Matrices via Local Transfer Learning*. Manuscript in preparation.

[2] **Yongzhuo Chen** and Sai Zhang. *Mapping Cell-Type-Specific Regulatory Variant Effects Associated with Amyotrophic Lateral Sclerosis Using MCformer*. Manuscript in preparation.

[3] **Yongzhuo Chen** and Michael Kane. *Prognostic Subtyping of Parkinson's Disease Using UK Biobank Accelerometry Data*. Manuscript in preparation.

Experience

Late-Fusion Multi-task Learning for Semiparametric Model

- Developed *adaptive late-fusion multi-task learning* to share estimates across heterogeneous studies without individual-level data exchange.
- Extended transfer to *potentially infinite-dimensional nuisance functions* and helped construct Neyman-orthogonal scores for semiparametric estimation.
- Established estimation-error bounds; nuisance transfer reduced median MSE by *59% (PLM) and 94% (CATE)* versus individual-task learning in reported simulations; applied the method to mammography-screening trials.

Experience (continued)

Adaptive Estimation and Inference for Time-Varying Transition and Precision Matrices via Local Transfer Learning

- Developed *adaptive local-transfer estimators* for time-varying transition and residual precision matrices, borrowing information across developmental stages.
- Derived estimation-error bounds and *bias-aware inference* for matrix entries; simulations showed shorter confidence intervals and coverage approaching nominal levels.
- Used optimal transport with longitudinal Wibj2 brain organoid data to identify changing cross-time gene relationships involving *Notch and Rho-GTPase pathways*.

Mapping Cell-Type-Specific Regulatory Variant Effects Associated with Amyotrophic Lateral Sclerosis Using MCformer

- Adapted AlphaGenome with LoRA for expression prediction across *20 motor-cortex cell populations*; held-out median Pearson correlations were *0.81 across genes* and *0.66 across cell types*.
- Predicted effects for *231,464,318 SNP–gene pairs*; reference–alternate expression differences agreed with matched eQTL effects (*Spearman* $r = 0.717$).
- Predicted enhancer–gene links with *AUPRC 0.47 versus 0.41* for a distance-only baseline on an astrocyte CRISPRi benchmark; applied S-LDSC and GO analysis to ALS-associated annotations.
- Developed a Bayesian framework to integrate cell-type-specific variant effects with genetic association statistics for candidate causal variant prioritization.

ALS Primary Motor Cortex Single-Nucleus Multiomic Atlas

- Applied scDRS to ALS survival-associated genetic signals in a collaborative single-nucleus RNA/ATAC atlas, identifying enrichment in *oligodendrocytes, astrocytes, and microglia*.

scGMP: Single-Cell Foundation Model–Guided Disease-Gene and Cell-Type Prioritization

- Integrated cell and gene embeddings from pretrained single-cell foundation models into an attention-based neural network to predict GWAS-associated disease genes.
- Used cell-level attention scores to prioritize disease-relevant cell types in Alzheimer's disease, highlighting microglia and perivascular macrophages.

Working Experience

Teaching Assistant — Data Science of Software System *Sep. 2023–Apr. 2024*

Department of Biostatistics, Yale University; Instructor: Michael Kane

- Supported instruction in programming and clinical-data analysis; graded assignments and exams and provided student assistance during office hours.

AT&T — Summer Intern, GraphRAG and Large Language Models *Summer 2024*

Supervisor: Elena Khusainova

- Built an LLM-driven pipeline to extract entities and relationships, construct domain-specific knowledge graphs, and store and query the graphs in *Neo4j*.
- Developed question-conditioned graph retrieval to supply relevant subgraph context to an LLM for specialized question answering and to reduce unsupported responses.
- Improved binary question-answering accuracy from *50% to 85%* (*+35 percentage points*) on the evaluated task.

Technical Skills

Statistical methods: Statistical inference, semiparametric estimation, double/debiased machine learning, transfer learning, time-varying models, precision matrices, optimal transport, S-LDSC.

Computational genomics: Single-cell RNA-seq, ATAC-seq, eQTL analysis, variant-effect prediction, GO enrichment, brain organoid analyses.

Programming and tools: Python, R, PyTorch, LaTeX, Neo4j, LLMs/GraphRAG, HPC/SLURM.